

Project Task D Pretty Printing

Up to now your compiler can already lex and parse input. In this project phase you will add a pretty printer to your compiler, i.e., functionality to generate nicely formatted source code from the parse tree. Therefore, you have to construct an abstract syntax tree (AST). Each AST node should be able to dump itself.

For a syntactically correct input program, `c4 --print-ast [file]` must print the output to stdout. This new switch must *augment* the `c4 --parse` functionality, i.e., its acceptance and rejection behavior as well as the respective return codes.

Download the example file `pretty.c` from Stud.IP. Invoking `c4 --print-ast pretty.c` should emit a char-exact copy of the source file. You can use the tool `diff` to compare your output to the reference file. Once again: *Your output must exactly match the reference output.*

You can use the pretty printer to debug your compiler. For example, the expression `a * b + d + e` should be dumped as `((a * b) + d) + e`. If you see a different output, your compiler handles associativity/operator precedence incorrectly.

In general, follow the reference file for placing newlines, spacing and so forth. The example mostly follows K&R style.¹ Additionally, consider the following guidelines for your pretty printer:

- Print everything in the original order of the source file.
- Use a tab character (`'\t'`) for one level of indentation; do not use spaces.
- Do not wrap long lines.
- Generally, for inner statements, start a newline and increment the indentation level for this inner statement.
- Extrapolate the placements of `else`-statements from the example.
- Do not indent an `if`-statement after an `else` and begin the `if`-statement on the same line.
- Each (sub-)*expression* and each (sub-)*declarator/abstract-declarator* must be fully parenthesized, except for atoms consisting of at most one token like `0`, `'c'`, `"abc"`, *identifiers* and so forth. In particular, original parentheses are discarded. Do not re-associate any subexpressions.
- The use of whitespace within expressions and statements can be extrapolated from the examples.
- The soft deadline for this milestone is 2025-12-21.
- Keep it simple!

¹see https://en.wikipedia.org/wiki/Indentation_style#K&R